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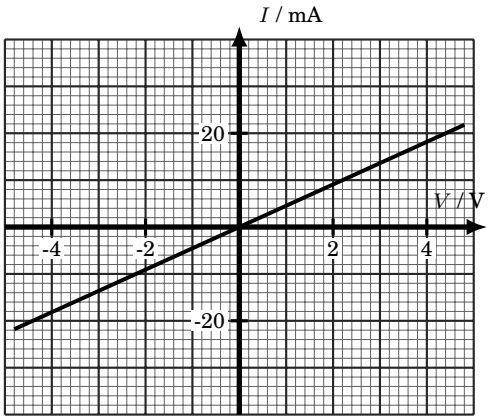
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Mr. Rawson • GCSE Physics

Ohmic and Non-Ohmic Conductors

Three components, three shapes. Read the shape and you know the component.

1 · Ohmic conductors resistance stays the same



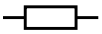
Graph A — a fixed resistor

The shape: a **straight line through the origin**.

Double the pd, and the current doubles too. So $R = \frac{V}{I}$ comes out **the same everywhere on the line**.

$$I \propto V$$

Examples



- a **fixed resistor**
- a **metal wire** — as long as it stays cool

The condition: *at a constant temperature.* Let it get hot and it stops being ohmic.

Worked example — the method

A point read off a *different* resistor's graph gives $V = 3.0\,\text{V}$ and $I = 25\,\text{mA}$.

Step 1 — convert the current to amps. $25\,\text{mA} = 25 \div 1000 = 0.025\,\text{A}$

Step 2 — divide. $R = \frac{V}{I} = \frac{3.0}{0.025} = \mathbf{120\,\Omega}$

Forgetting step 1 gives 0.12 — a thousand times too small. The current axis on this sheet is always in **milliamps**.

Finding resistance from any graph

Pick a point on the line. Read V across. Read I up. Divide. $R = \frac{V}{I}$ — this works on *every* graph on this sheet.

The check: do it again at a second point. Same answer → the resistance is **constant** (ohmic). Different answer → the resistance has **changed**.

Exam-style question [4 marks]

Graph A is a real fixed resistor. Use the graph to find its resistance.

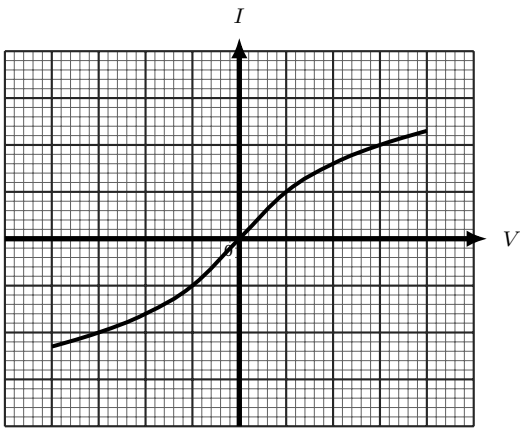
Choose a point where the line crosses a grid line *exactly* — it makes the reading easier.

The point I read: $V =$ _____ V $\quad I =$ _____ mA

Any point on the line gives the same answer. Reading $V = 4.4\,V$ and $I = 20\,mA$: $20\,mA = 0.020\,A$, so $R = \frac{4.4}{0.020} = \mathbf{220\,\Omega}$. (Or $V = 2.2\,V$, $I = 10\,mA \Rightarrow 2.2/0.010 = 220\,\Omega$.) 1 mark a valid pair of readings, 1 for mA → A, 1 for the division, 1 for $220\,\Omega$ with the unit.

2 · Non-ohmic conductors resistance changes as you use it

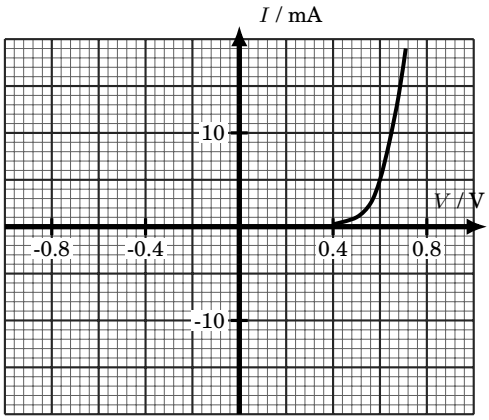
The resistance of a **lamp**, **diode**, **thermistor** or **LDR** is **not** constant — so their graphs are not straight lines.



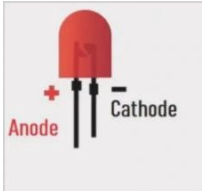
Graph B



the filament and its support wires



Graph C — a diode
note the scale: 1 square = 0.2 V



an LED is a diode — it only works one way round

2a The filament lamp

The shape: bends over towards the V axis — the same way on *both* sides.

Why: bigger current → filament **hotter** → ions **vibrate more** → more **collisions** → **resistance increases**.

Examples: a torch bulb, a car headlamp, an old-style light bulb.

Exam-style question [3 marks]

Explain why the resistance of a filament lamp increases as the potential difference across it increases.

A larger pd gives a larger current, which makes the filament hotter. The metal ions vibrate more, so the electrons collide with them more often, and the resistance increases.

2b The diode

The shape: flat along the axis one way, then rising steeply the other way.

Why: a diode lets current through in **one direction only**. Forwards, it conducts once the pd passes about 0.6 V. Backwards, its resistance is **very high**, so the current is almost zero.

Examples: an **LED** is a diode — connect it the wrong way round and it simply will not light. Also the charging circuit in a phone, a solar panel, and anything that must not be connected backwards.

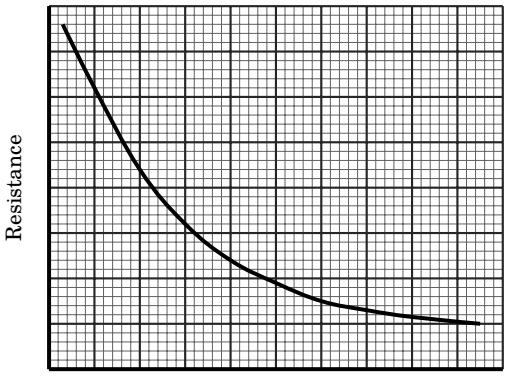
Exam-style question [2 marks]

A diode is connected in a circuit with a cell and a lamp, and the lamp lights. The diode is then turned round. Describe and explain what happens to the lamp.

The lamp goes out. In the reverse direction the diode has a very high resistance, so there is almost no current in the circuit.

Name: _____ **Date:** _____

Both components below are non-ohmic too — but their graphs use **different axes**. Instead of I against V, we plot **resistance** against the thing the component responds to.

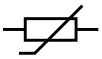


Graph D



the black bead at the top is the temperature sensor

2c The thermistor



The shape: resistance **falls** as the temperature **rises** — steeply at first, then levelling off.

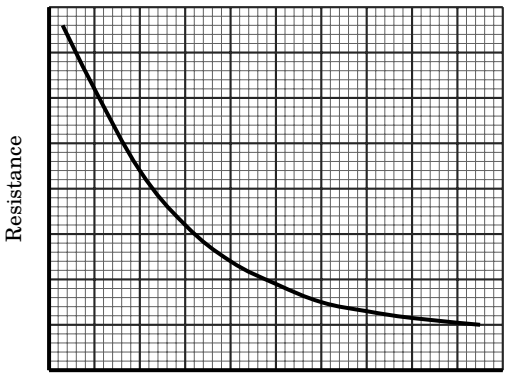
Remember it as: *hotter → lower resistance → bigger current.*

Examples: a thermostat, a fire alarm, the temperature sensor in an oven or a car engine.

Exam-style question [3 marks]

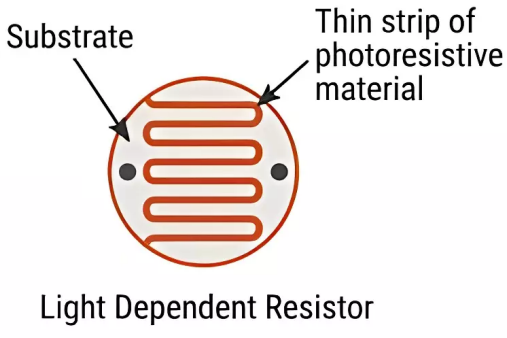
A thermistor is connected in series with a lamp and a cell. Describe and explain what happens to the brightness of the lamp as the room gets warmer.

The lamp gets brighter. As the temperature rises the thermistor's resistance decreases, so the total resistance of the circuit falls and the current increases.



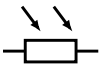
Graph E





the track is the part that responds to light

2d The LDR



LDR stands for **light dependent resistor**.

The shape: the *same* falling curve as the thermistor — but against **light intensity**. Resistance **falls** as it gets **brighter**.

Remember it as: *brighter → lower resistance → bigger current.*

Examples: street lights that come on at dusk, automatic camera exposure, a phone screen that dims in a dark room.

Exam-style question [3 marks]

Explain how an LDR can be used to switch a street light on automatically when it gets dark.

As the light level falls, the resistance of the LDR increases. That increases the share of the potential difference across the LDR, and the circuit uses this change to switch the street light on.

Name: _____ **Date:** _____

3 · Questions graphs A, B and C are on the front

1. State what is meant by an **ohmic conductor**. **[2]**

   A conductor whose resistance stays constant (at constant temperature), so the current through it is directly proportional to the pd across it.

2. Name the component that produced each graph. **[3]**

   A: *fixed resistor* **B:** *filament lamp* **C:** *diode*

3. Use **graph B** to find the resistance of the lamp at $V = 1.0\,\text{V}$ and at $V = 3.0\,\text{V}$. Show your working, and say what has happened to the resistance. **[4]**

   At $V = 1.0\,V$, $I = 0.20\,A$: $R = V/I = 1.0/0.20 = \mathbf{5.0}\,\Omega$.
   At $V = 3.0\,V$, $I = 0.40\,A$: $R = V/I = 3.0/0.40 = \mathbf{7.5}\,\Omega$.
   *The resistance has **increased**, because the filament is hotter at the higher pd.* Accept readings within one small square.

4. Jamie found the resistance for **graph A** by measuring its gradient, and wants to use the same method on **graph B**. State what Jamie should do instead for graph B, and explain why the method that worked on graph A does not carry across. **[3]**

   *He should read **one point** off the curve and calculate $R = V/I$ at that point. Graph A is a straight line through the origin, so its gradient equals $1/R$ and happens to give the right answer there — but graph B is a curve, so it has no single gradient, and resistance is V/I rather than $\Delta I/\Delta V$ in any case. $R = V/I$ at a point works for both.*

5. Explain, in terms of the filament, **why** graph B bends. **[3]**

   As the current increases the filament gets hotter. The metal ions vibrate more, so the electrons collide with them more often, and the resistance increases — so each extra volt gives less extra current.

6. Look at the **left-hand half** of the graphs, where the pd has been reversed. **[4]**

   (a) Describe what **graph C** does, and explain why.

   Almost no current flows — the line lies flat along the V axis. A diode conducts in one direction only; in reverse its resistance is very high.

   (b) Graph B is symmetrical about the origin. What does that tell you about a filament lamp?

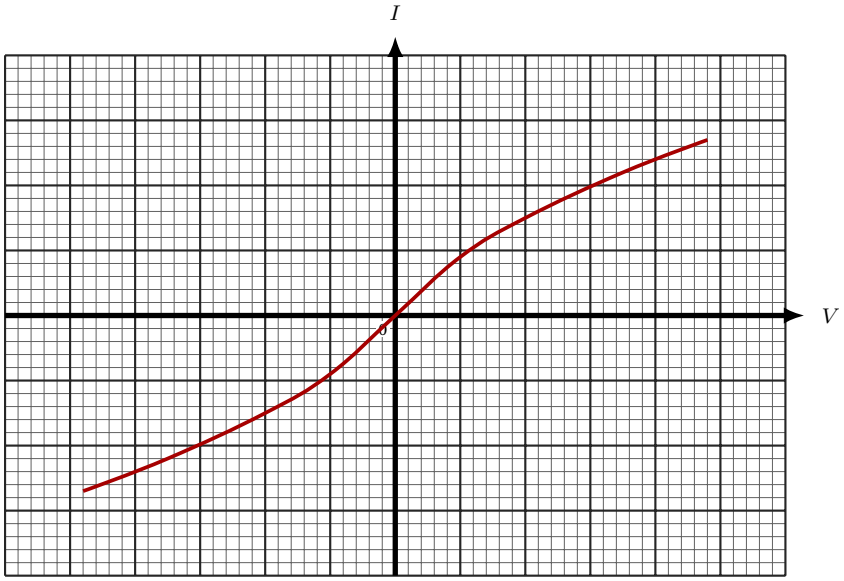
   It behaves the same way whichever direction the current flows. It is non-ohmic, but it is not one-way.

7. A fixed resistor is left switched on until it becomes noticeably warm, and the test is repeated. Circle what happens to its graph, and explain in one sentence. **[2]**

   it begins to bend over it stays perfectly straight it flattens along the axis

   It is no longer at a constant temperature, so its resistance is no longer constant — the line starts to curve and it stops behaving ohmically.

8. Sketch the graph you would expect for a **filament lamp**. Use both halves of the axes. **[3]**



Through the origin; curving over towards the V axis; drawn in both halves. One mark each.

Filament lamp photograph: KMJ via Wikimedia Commons, CC BY-SA 3.0.